Hongtao Hao

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Summary

Research Scientist at Meta building deep-learning ranking models for ads. Ph.D. in Computer Sciences (2026) in machine learning for healthcare: Bayesian and Transformer models of disease progression, and an LLM-assisted agentic workflow for diagnosis.

Experience

Meta Platforms, Inc. Bellevue, WA
Research Scientist, Ads Ranking *Jul 2026 – Present*

- Develop deep-learning ranking models and multi-armed bandit algorithms to decide how an ad is presented, including which creative format to show.
- Improve advertising models end to end: model development, large-scale GPU training, and evaluation.

University of Wisconsin–Madison Madison, WI
Graduate Researcher, Machine Learning for Healthcare *Aug 2021 – May 2026*

- Designed five probabilistic and deep-learning models (SA-EBM, BEBM, JPM, TEMPO, SUBTRACE) that infer disease event order, patient stage, subtypes, and timelines from cross-sectional biomarker data; applied them to Alzheimer's cohorts (ADNI, NACC) and released them as open-source Python packages.
- **TEMPO** (Transformer, simulation-based training) vs. prior state of the art on simulated benchmarks: 53–59% fewer misordered events, 25–61% lower staging error, sub-second inference vs. 5–55 minutes.
- **BEBM** (Bayesian subtype model) vs. SuStaIn, the most widely used method: 27% better ordering, 89% better staging, 56% better subtype assignment, and faster.
- Ran 9,000+ distributed experiments on HTCondor clusters. Co-developed an LLM diagnostic framework that pairs LLM-generated questions with Bayesian belief updates to weigh information against cost (COLM 2026 workshop).

Bosch Research North America Sunnyvale, CA
Research Intern, LLMs and Human-AI Interaction *Jun 2023 – Aug 2023*

- Built a full-stack LLM data-analysis application (Python, MongoDB, Streamlit) with an automated evaluation pipeline (LangChain, Microsoft Azure); contributed to CHI 2024 EA and IUI 2024.

Indiana University, YY Lab Bloomington, IN
Research Assistant, Data Visualization *Aug 2020 – May 2021*

- Co-created a COVID-19 trend dashboard (D3.js, Observable) named among Observable's Top 10 Liked notebooks of 2020; automated daily synchronization with Our World in Data for 200+ countries and regions.

Skills

- **ML:** Transformers, ranking models, multi-armed bandits, Bayesian inference/MCMC, time series; PyTorch, TensorFlow, scikit-learn, XGBoost.
- **Generative AI:** LLM applications, RAG, LangChain, LoRA/PEFT fine-tuning, RLHF, DPO, GRPO, LLM evaluation.
- **Engineering:** Python packages, SQL, TypeScript/React/Next.js, FastAPI, Docker, CI/CD; HTCondor/Slurm, GCP, Azure.

Selected Publications *(first author unless noted)*

- TEMPO: Transformers for Temporal Disease Progression from Cross-Sectional Data. *CHIL 2026*.
- Adaptive Bayesian Active Querying with LLMs for Efficient Information Gathering. *DAIH @ COLM 2026* (second author).
- Bayesian Event-Based Model for Disease Subtype and Stage Inference. *ML4H 2025*; NeurIPS 2025 Workshop.
- Joint Progression Modeling (JPM): A Probabilistic Framework for Mixed-Pathology Progression. *ML4H 2025*; NeurIPS 2025 Workshop.
- Stage-Aware Event-Based Modeling (SA-EBM) for Disease Progression. *MLHC 2025*.
- Investigating Interaction Modes and User Agency in Human-LLM Collaboration. *CHI 2024 EA* (fourth author).

Education

Ph.D., Computer Sciences, University of Wisconsin–Madison, 2026. Advisor: Joseph L. Austerweil.

Service

Reviewer for NeurIPS, ML4H, CHIL, IEEE VIS, IUI, and BMC MIDM; program committee, DAIH @ COLM 2026.